# SVARA

**Sing. Learn. Find Your Voice.**

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# THE PROBLEM

Karaoke, feedback, learning, and progress often live in separate experiences.

01 **Limited Karaoke Customization**

02 **Lack of Personalized Music Learning**



03 **Inadequate Real-Time Singing Feedback**

04 **Subjective Voice Assessment**

# One product. Four connected modules.

01

## Karaoke

Import, separate, mix, listen, export

02

## Learn

Voice exercises with Carnatic elements

03

## Voice Analysis

Pitch, breath, and range detection

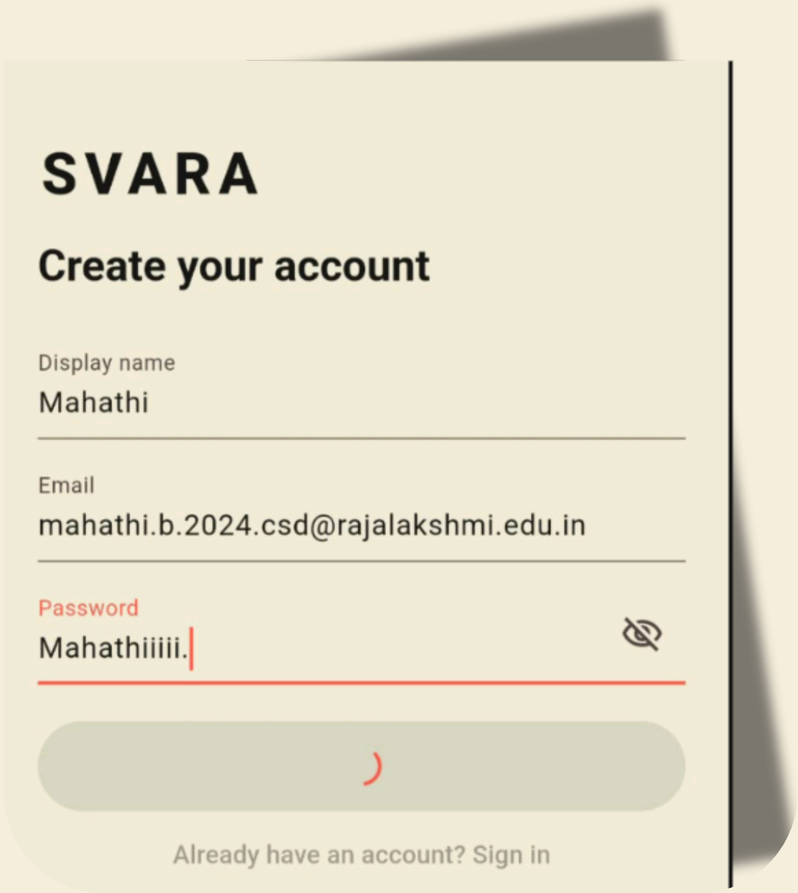
04

## Vocal Profile

Personal metrics and progress

# From sign-up to a personalised start.

1. **Sign-up with email**
2. **Supabase email verification**
3. **Sign in**
4. **Vocal profile setup**




Illustrative UI / mock data

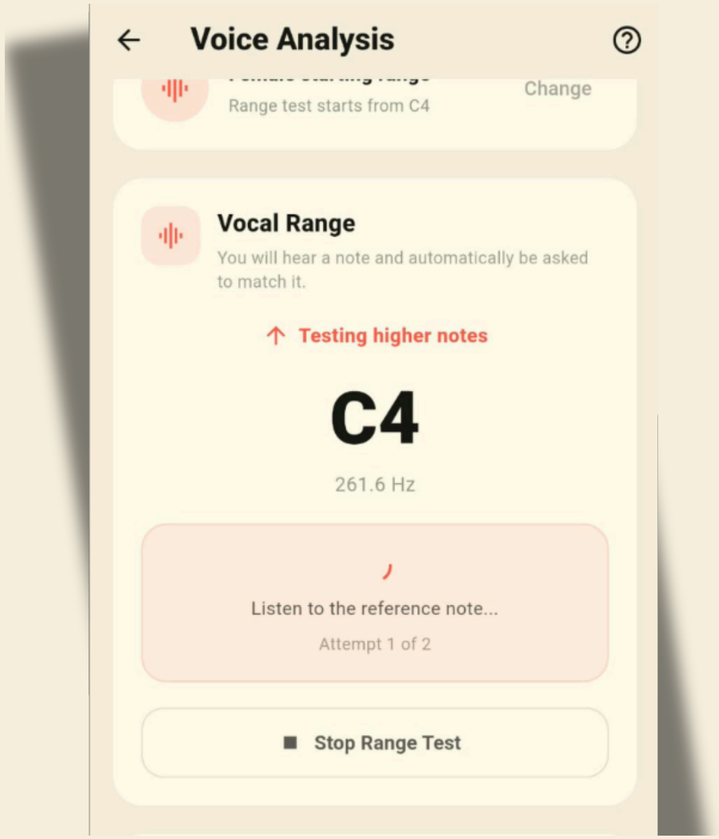
# Vocal range, tested note by note.

C3/C4 are selectable starting references

**START** **Choose your starting range (C3 or C4)**

**UP** **Chromatic test upward · 2 attempts per note**

**DOWN** **Test downward from C3 for lower limit**



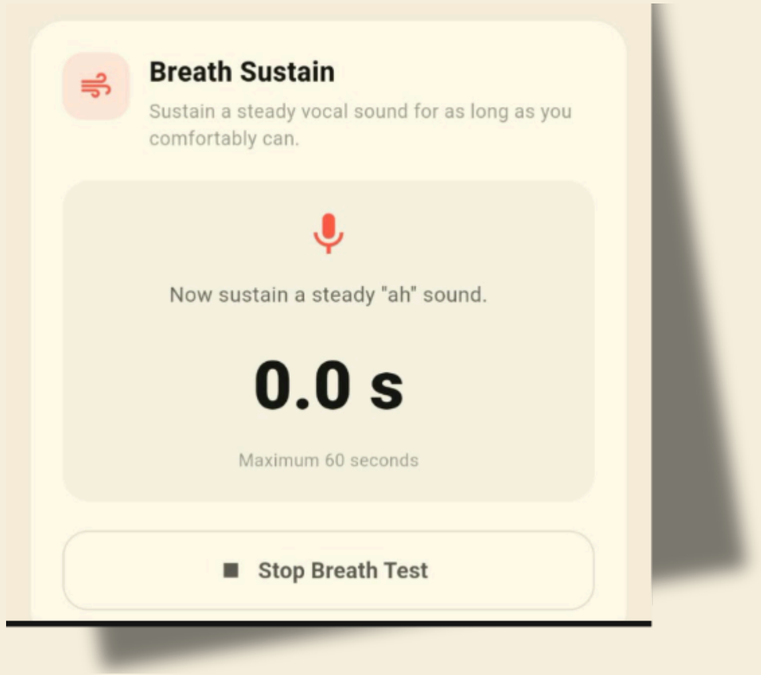

Illustrative UI / mock data

# Breath capacity, approximated through sustained sound.

The test records phonation duration as a practical proxy — not a direct lung-volume measurement.

# Sustain.
# Measure.
# Reflect.

Hold a comfortable reference note. SVARA captures the duration and waveform for the session summary.




**Illustrative UI / mock data**

# Karaoke Studio puts every stem within reach.

Import MP3 → separate stems → balance each layer → listen and export.

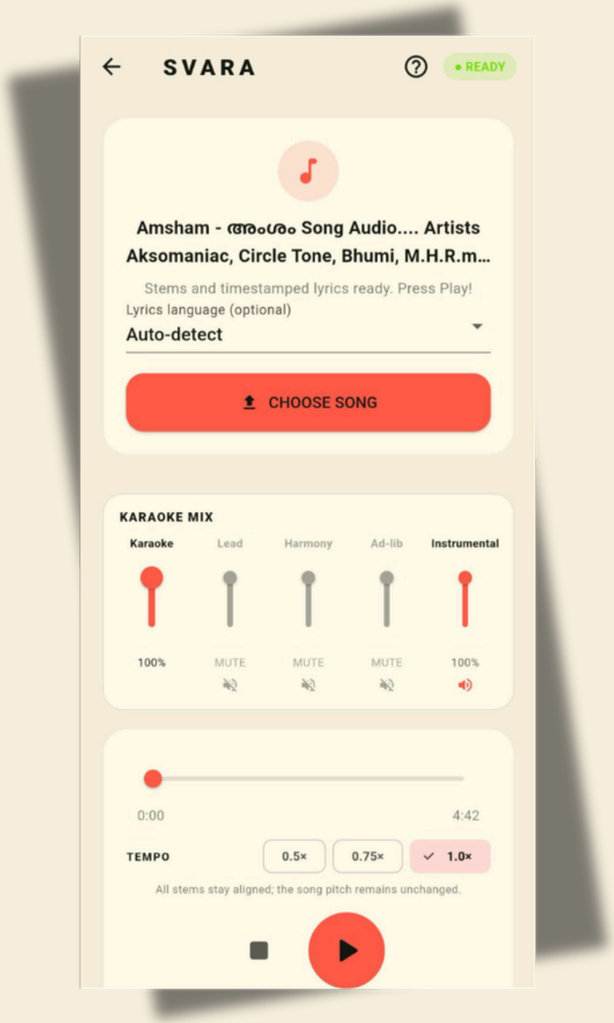
**Instrumental / karaoke**

**Backing vocals**

**Lead vocal**

**Ad-libs**

**Independent volume mix · mute / solo · preview · export**




Illustrative UI / mock data

# Learn and Improve

A flexible practice space for voice types, warm up companion.

## Chest voice

Grounded resonance

## Head voice

Light upper register

## Mixed voice

Balanced transition

## Tala

Rhythmic cycles

## Arpeggios

Pitch movement

## learn a song

Pitch + tempo

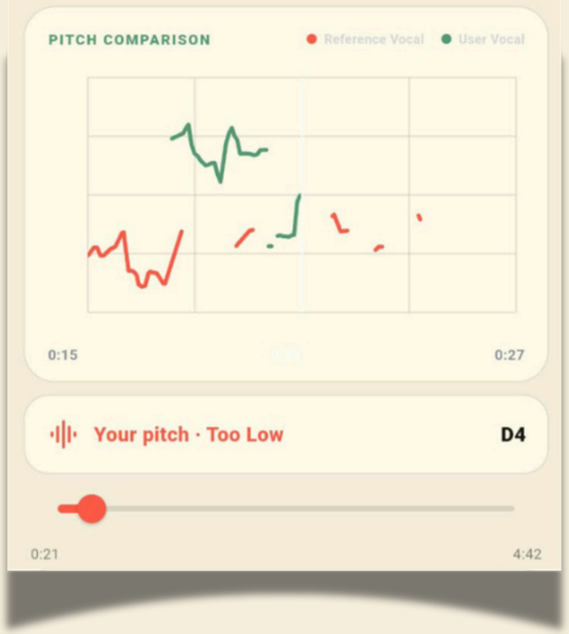
# A complete learn-a-song loop.

Proposed or intended — verify against implementation. Proposed features are not automatically verified as implemented.

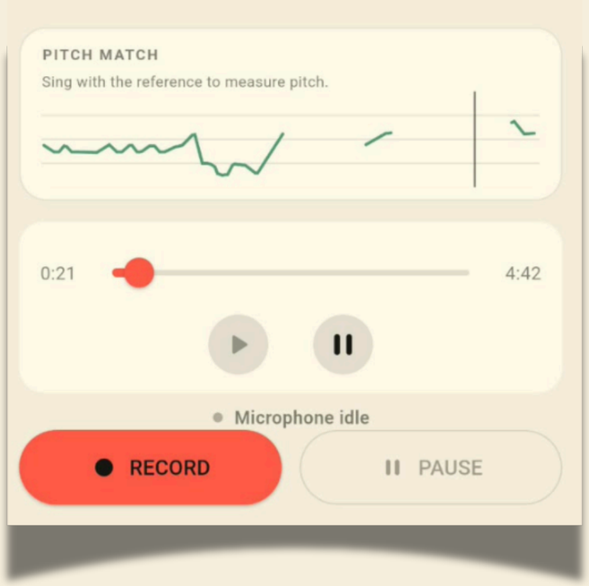
**1 Separate vocal stem**




**2 Sing + record**

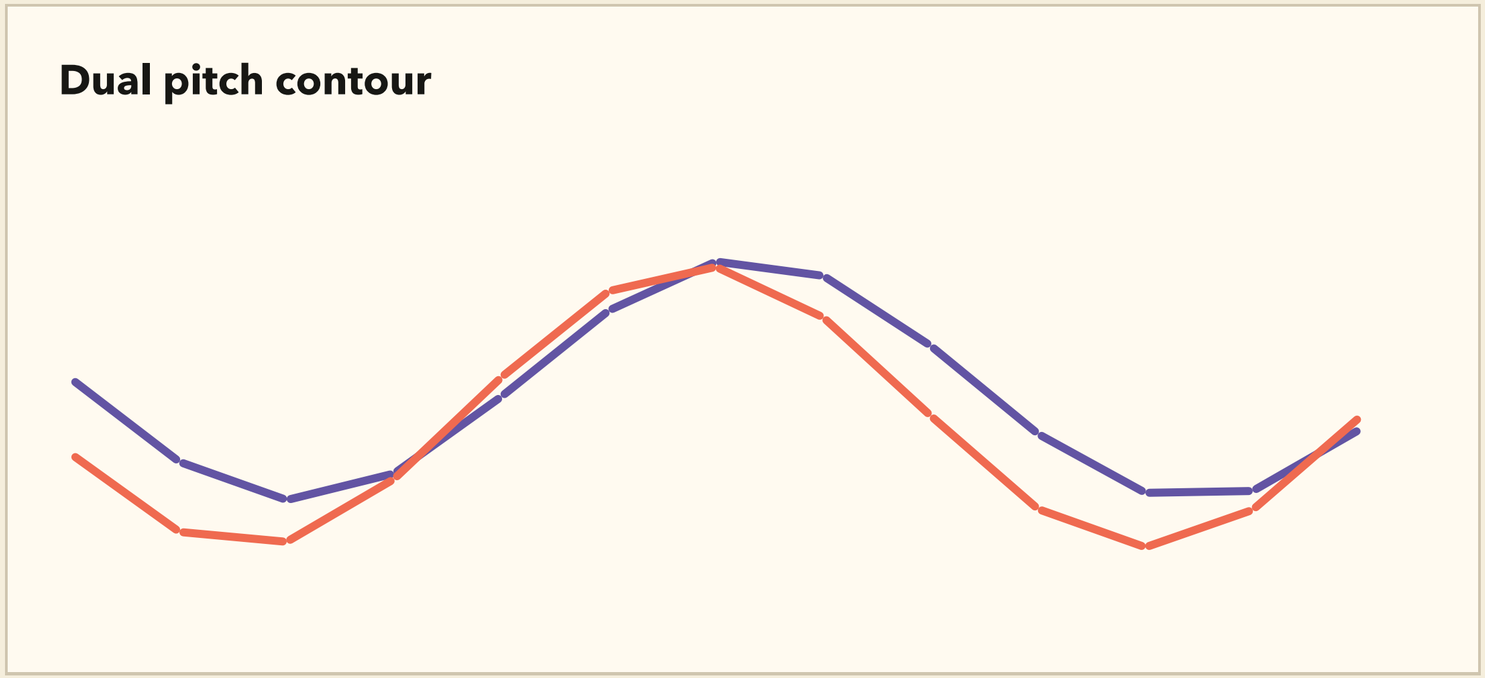



**3 Align pitch contours**



# Feedback at the exact moment it matters.

Illustrative UI / mock data. Stability and timing appear only if measured by the implementation.




**Sharp / flat**

Timestamped markers

**Stability + timing**

When measured

**Practice controls**

Loop · slow · re-record · download

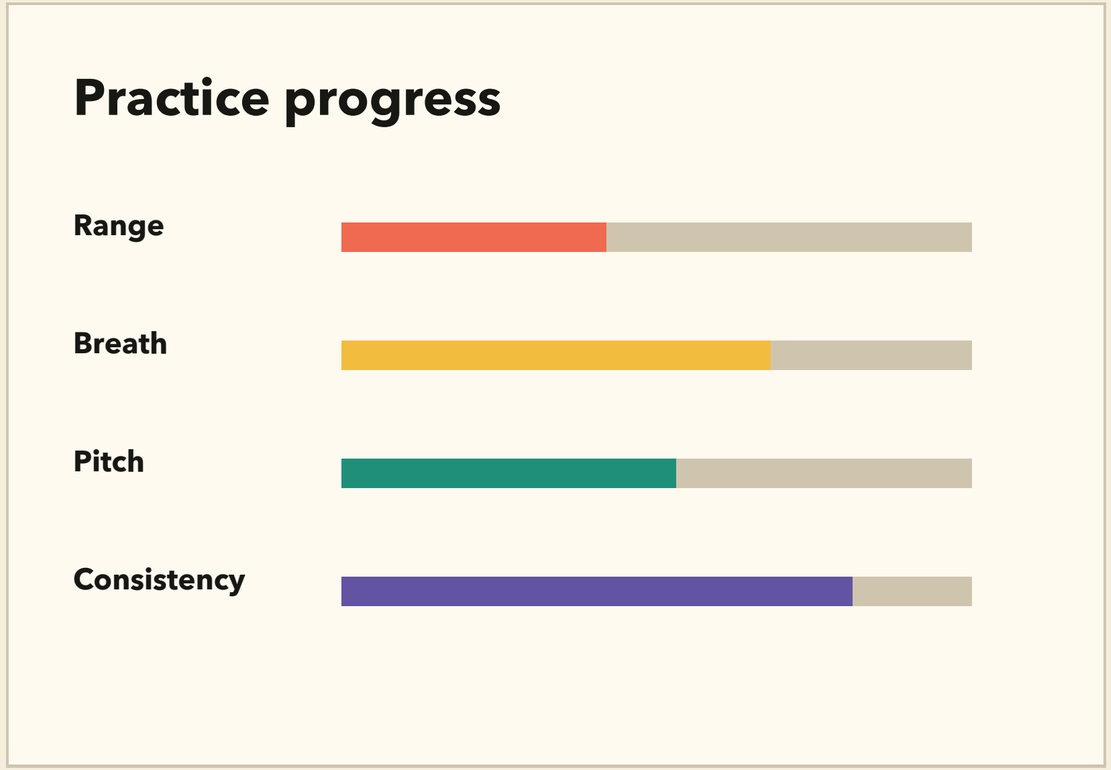
# Your work, organised into a visible practice history.

**Exported audio** — MP3s · stems · mixes

**Practice log** — Daily time entries

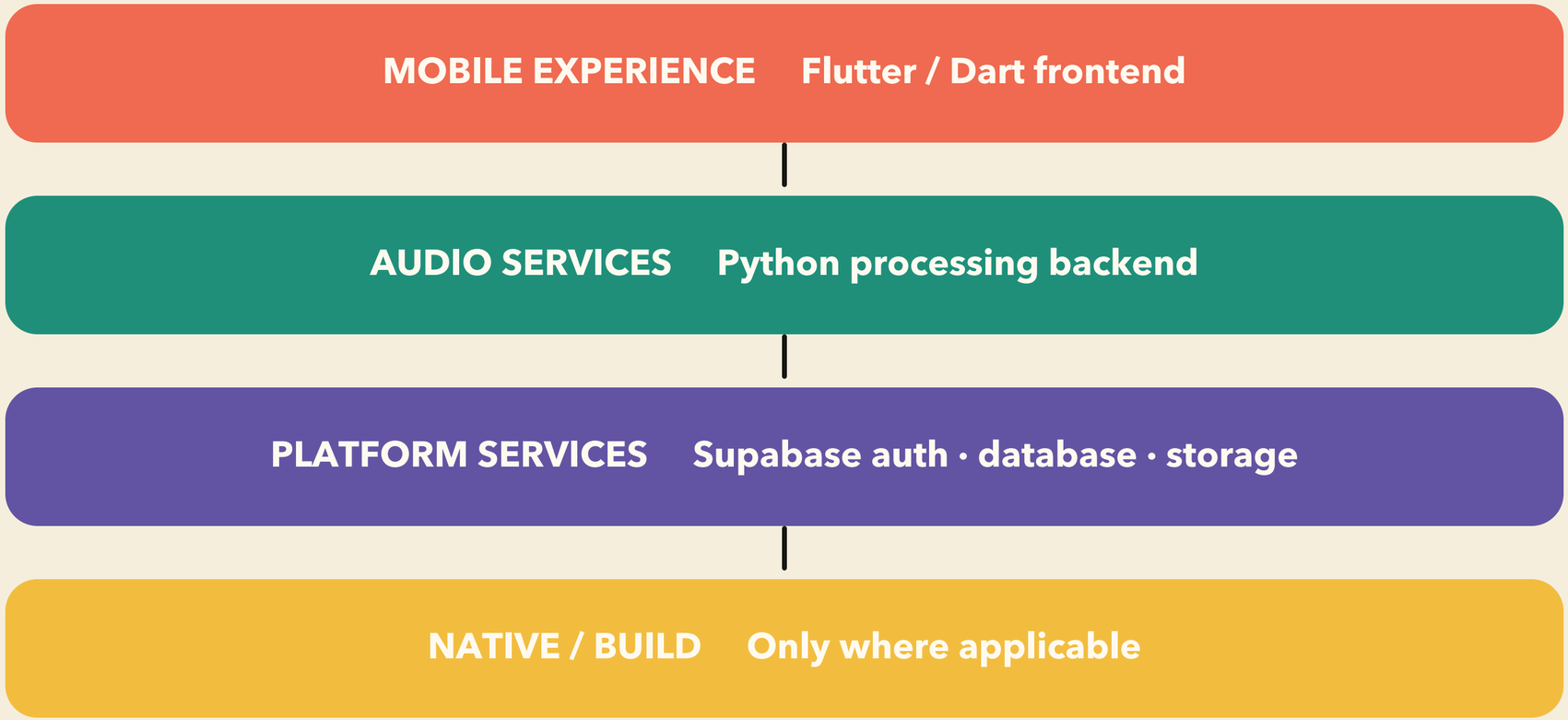
**Activity history** — Songs · lessons

**Saved reports** — Analysis snapshots



# A layered, explainable architecture.

Implementation details are described directly — not inferred from GitHub language statistics.



# Clear about what exists — and what comes next.

Proposed or intended functionality is not automatically verified as implemented.

**IMPLEMENTED**

## Integrated practice journey

Core connected flow represented in the project.

**PARTIAL**

## Measurable evaluation plan

Evaluation design exists; completion and evidence may vary.

**PROPOSED**

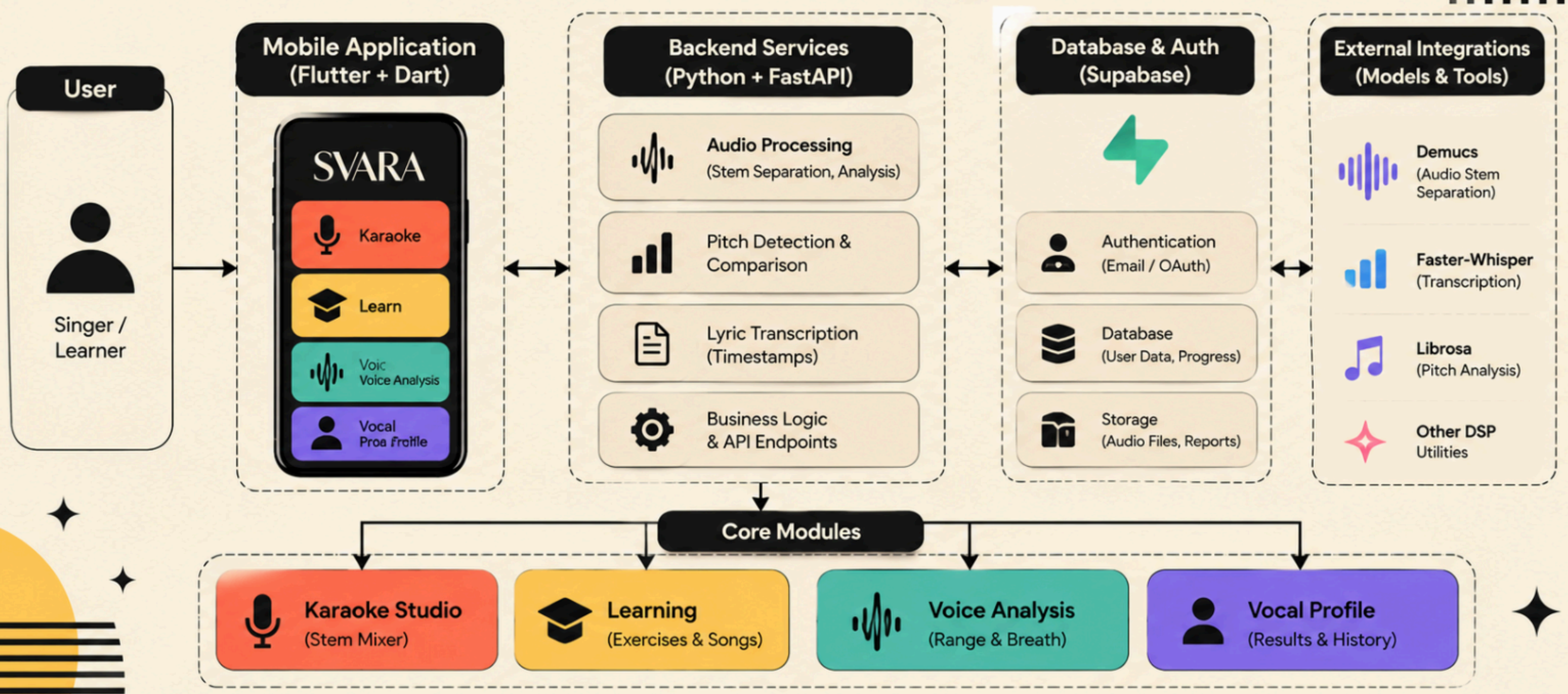
## Feature completion + expansion

Future work requires implementation and verification.

**Limitations are acknowledged; status labels must be checked against the current build and project evidence.**

# SVARA | System Architecture

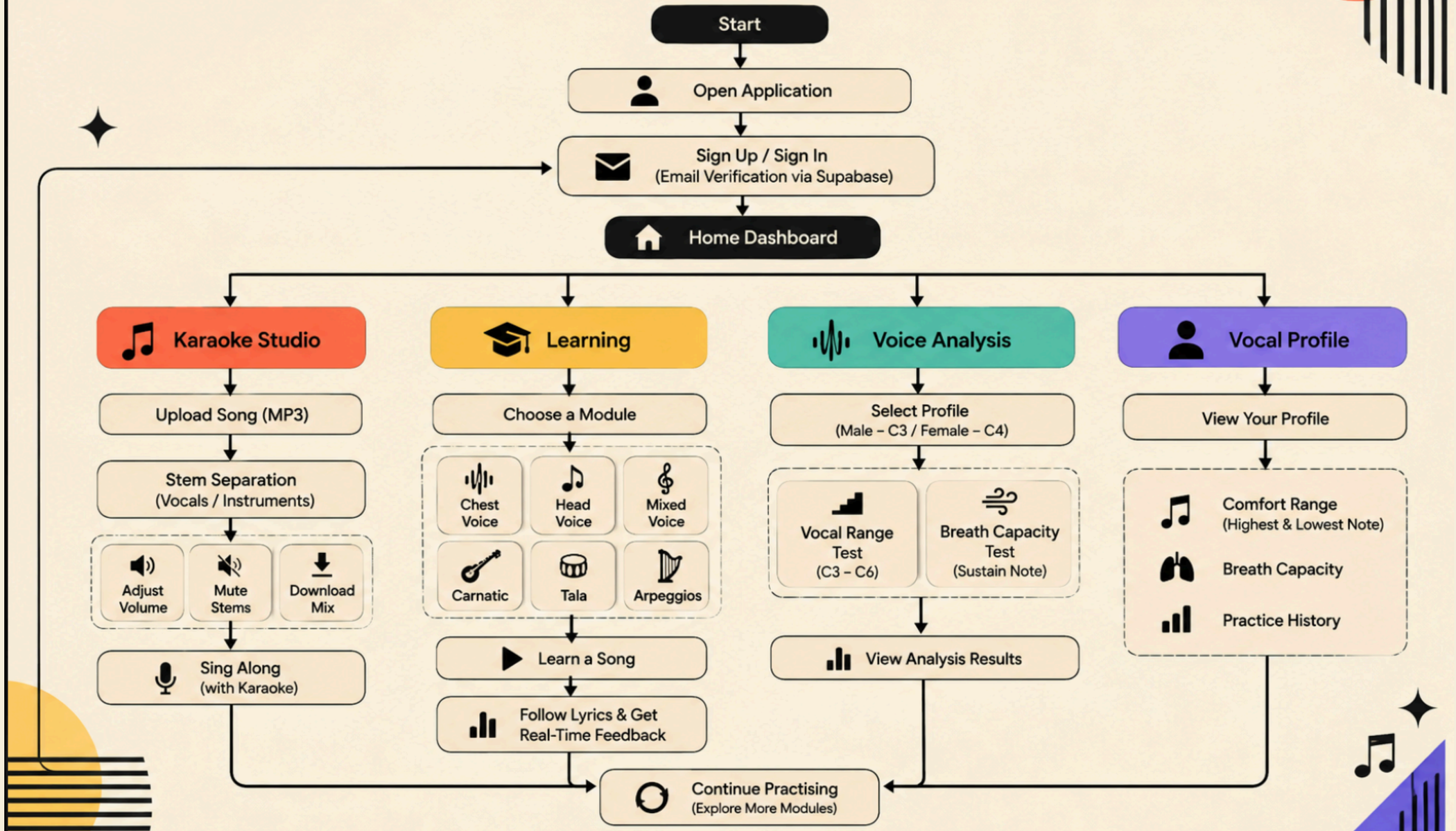
*An integrated platform for vocal learning*



# SVARA

## Application Flow Diagram

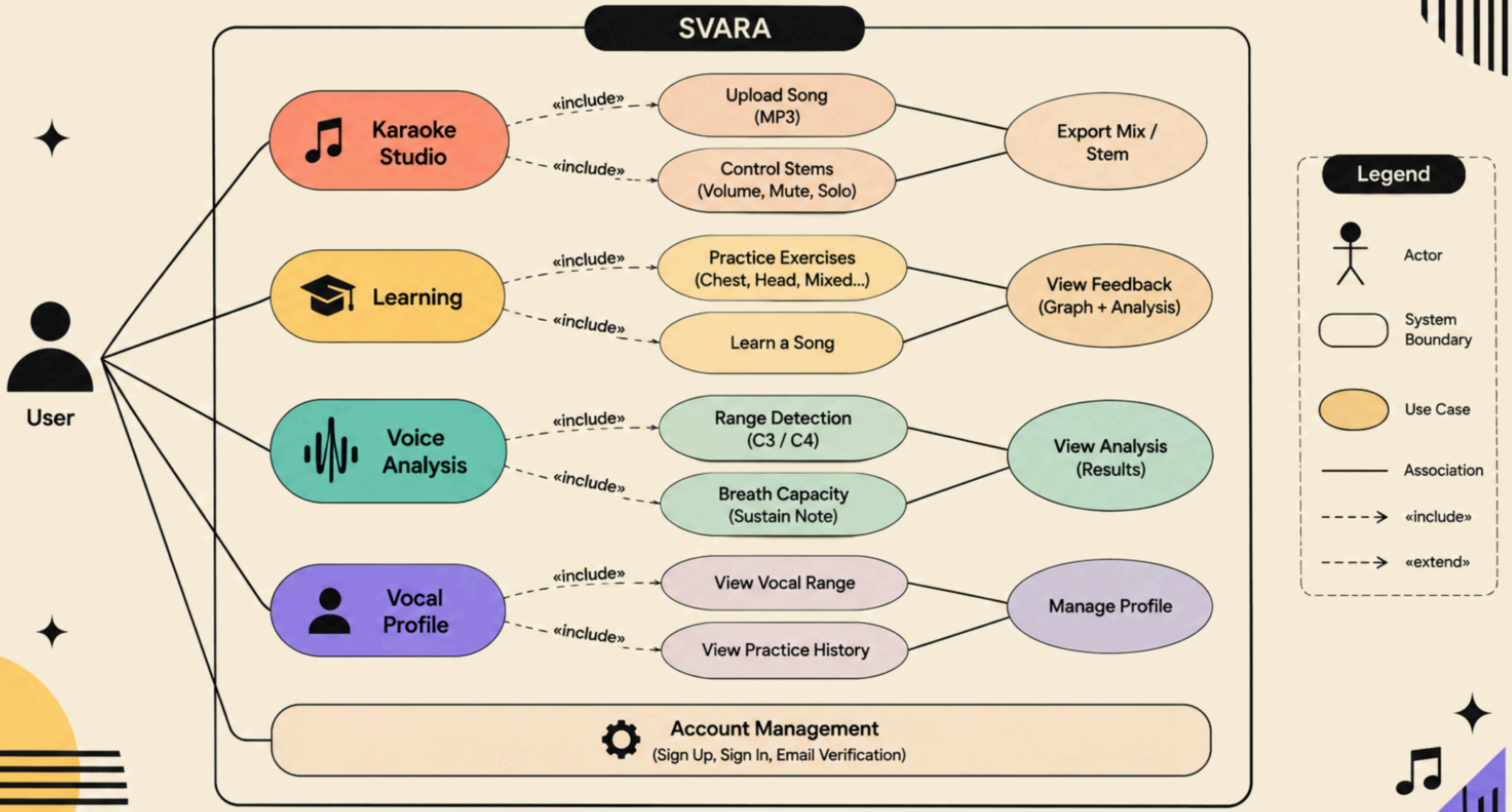
*Sing. Learn. Find your voice.*



# SVARA

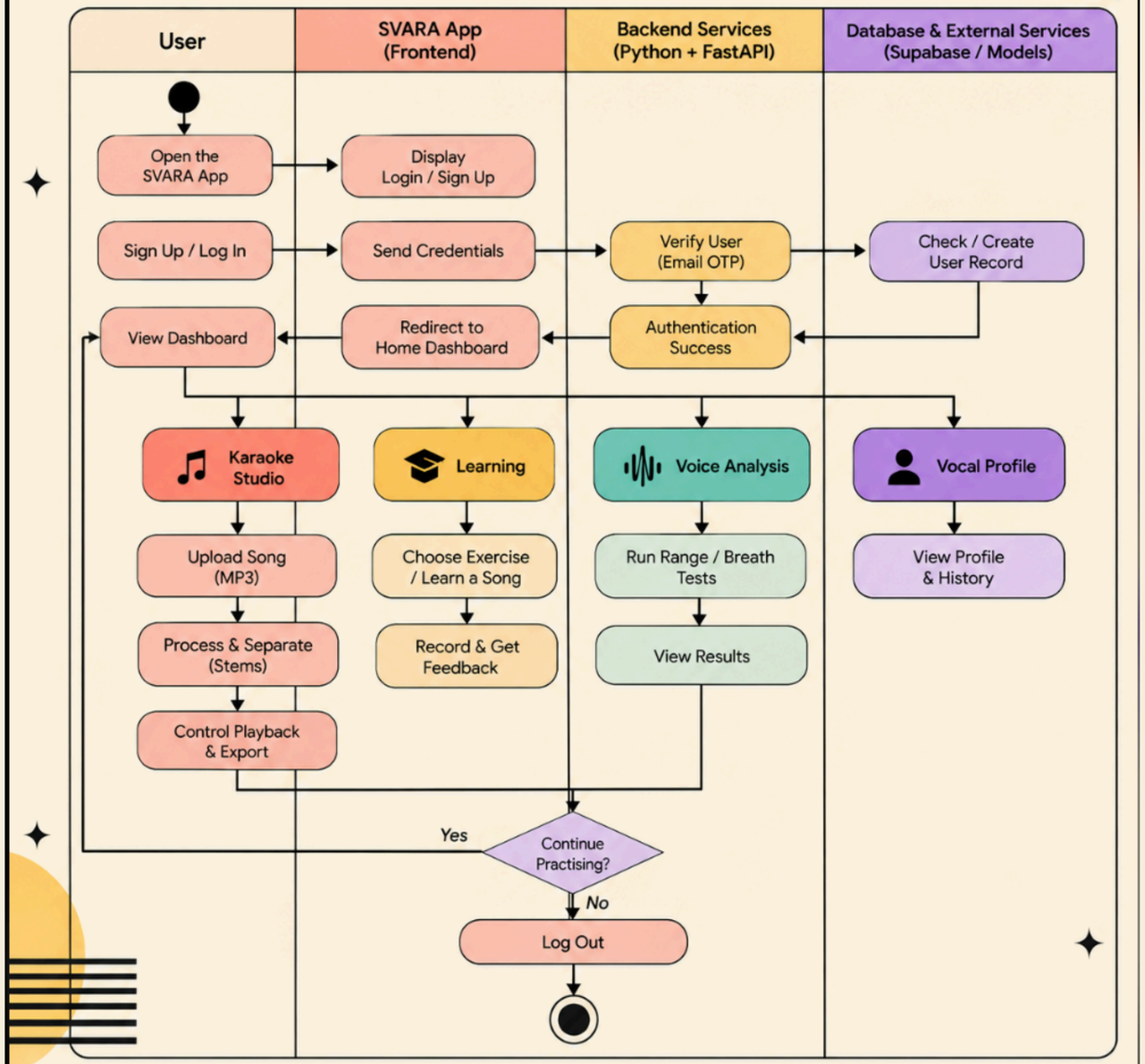
## Use Case Diagram

*Sing. Learn. Find your voice.*



# SVARA | Activity Diagram

*Flow of actions within the SVARA application*



# SVARA

**Sing. Learn. Find Your Voice.**

Thank you